

REGIONAL LOGISTICS REPORT

Analysis of Late Delivery Rates, Review Scores and Seller → Customer Routes

1. Executive Summary

This regional analysis evaluates whether delivery performance and customer satisfaction vary systematically across customer states and seller-state → customer-state routes. The analysis combines order, customer, seller, review and order-item information into a master dataset and focuses on delivered orders. After removing duplicate order records at the order level, 96,476 unique delivered orders were retained for the core state-level analysis.

Key findings: Alagoas (AL) has the highest late-delivery rate among statistically reliable states at 21.4%, followed by Maranhão (MA) at 17.4% and Sergipe (SE) at 15.2%. These same states also appear near the bottom in average review score, with MA at 3.83 and AL at 3.86. At route level, PR → AL is the strongest late-delivery hotspot at 32.6%, while PR → AL also records the lowest route review score at 3.44. These results indicate a meaningful relationship between poor delivery outcomes and weaker customer satisfaction, although the analysis is descriptive and does not establish causality.

2. Data & Methodology

Data preparation. The analysis uses customer, orders, order-items, order-reviews and seller datasets. Relevant date fields were converted to datetime format, after which the tables were merged using order_id, customer_id and seller_id. The resulting master table contains 114,092 rows and 14 variables before delivery filtering.

Delivery metrics. Only orders with a recorded customer-delivery date were retained. Delivery time was calculated as the number of days between purchase and customer delivery. Delay was calculated as actual delivery date minus estimated delivery date; positive values indicate late delivery and negative values indicate delivery before the estimate. An order was classified as late when delay_days > 0.

Reliability controls. For state-level comparisons, states with fewer than 100 orders were excluded, leaving 24 of 27 states. For route-level comparisons, routes with fewer than 30 orders were excluded. Duplicate order_id records were removed before the main state-level metrics to avoid double counting.

3. Regional Performance Findings

State	Orders	Late Delivery	Avg. Review
AL	397	21.4%	3.86
MA	717	17.4%	3.83
SE	335	15.2%	3.91
PI	476	13.9%	3.99
CE	1,279	13.8%	3.94
BA	3,256	12.2%	3.93
RJ	12,353	12.1%	3.96

Interpretation. The strongest concentration of delivery risk is visible in AL, MA and SE. Importantly, these states also have relatively low review scores, supporting the project hypothesis that regions experiencing poorer delivery outcomes tend to have weaker customer satisfaction. However, state size varies substantially, so the results should be interpreted together with order volume.

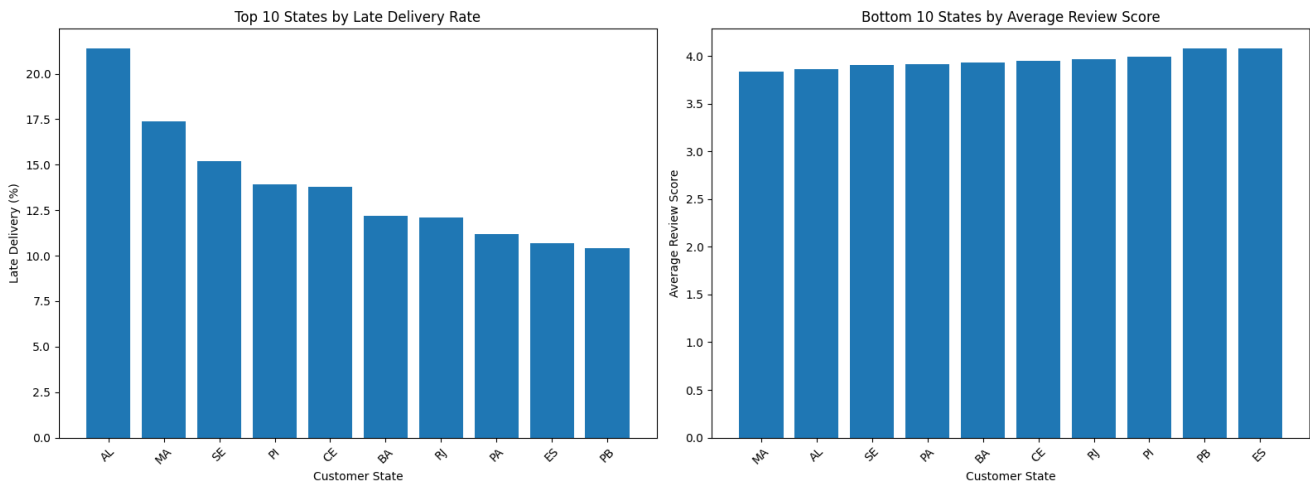


Figure 1. Top 10 customer states by late-delivery rate and bottom 10 by average review score.

4. Geographic View of Delivery and Satisfaction

The state-level maps reinforce the numerical findings by showing how performance varies geographically. The late-delivery map highlights substantial variation across customer states, while the review-score map shows that customer satisfaction also differs by region. The darkest/highest-risk areas in the late-delivery view are concentrated in several northern and northeastern destinations, whereas higher review scores are more visible in states such as SP, PR, RS and MG.

States by Late Delivery Rate

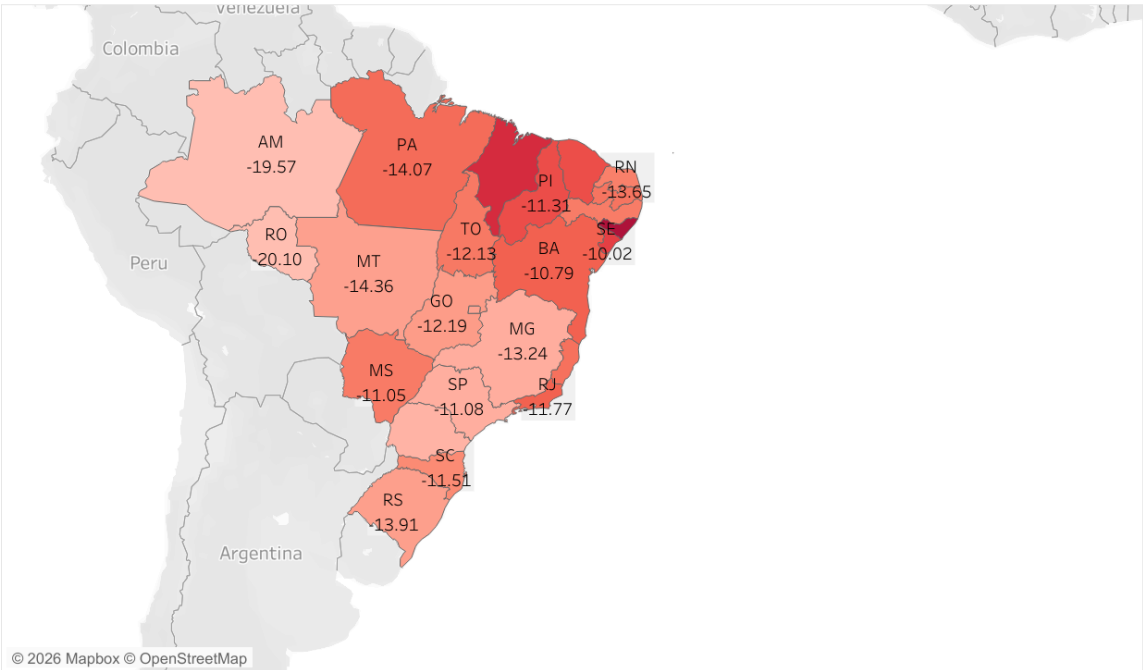


Figure 2. States by late-delivery rate. Values shown on the map are average delay days.

States by Average Review Score

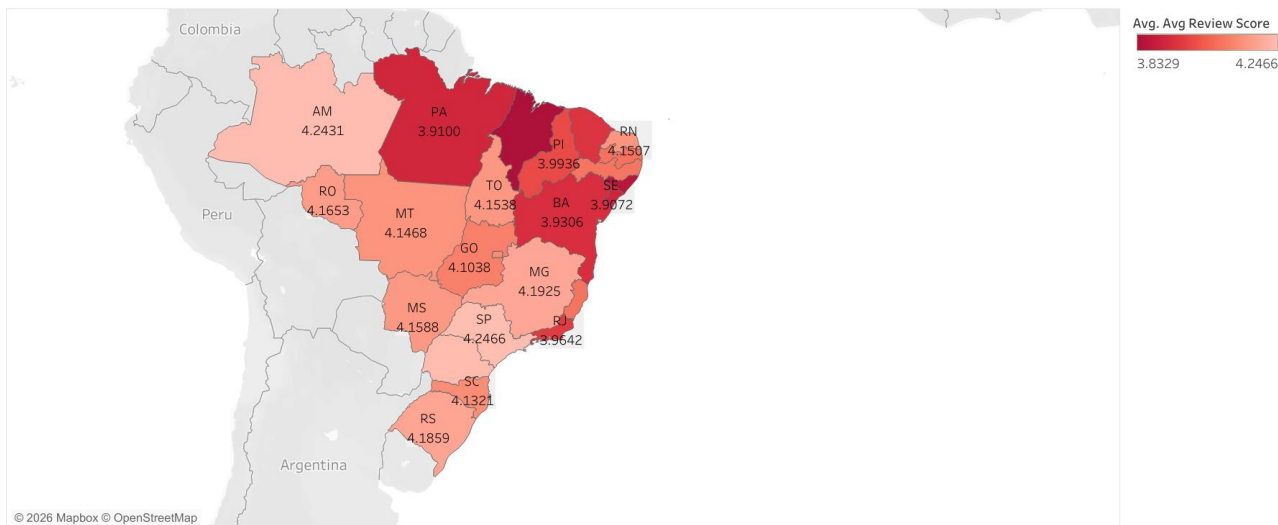


Figure 3. States by average review score.

5. What the Regional Pattern Suggests

- AL is the clearest state-level priority: it combines the highest reliable late-delivery rate (21.4%) with a low average review score (3.86).
- MA has the lowest average review score among the reliable states (3.83) and the second-highest late-delivery rate (17.4%).
- SE and PI also show elevated late-delivery rates (15.2% and 13.9%), while their review scores remain below the stronger-performing states.
- Large states such as RJ and BA matter operationally because even moderate late-delivery rates affect a large number of orders.
- Some states with high late-delivery percentages have relatively small order volumes; the project's minimum-order filtering appropriately reduces the risk of over interpreting very small samples.

Important metric nuance. Average delay_days is negative across the reported states because early deliveries offset late deliveries when averaged across all delivered orders. Therefore, a negative average delay does not mean that late deliveries are absent. The percentage of orders classified as late is the more direct indicator of late-delivery incidence.

6. Route-Level Performance Analysis

Route analysis examines whether specific seller-state → customer-state combinations create concentrated delivery or satisfaction problems. Routes were evaluated only when they contained at least 30 orders. This threshold helps balance hotspot detection with statistical reliability.

Route	Orders	Late %	Review	Interpretation
PR → AL	36	32.6%	3.44	Highest late rate and lowest review score
RJ → CE	54	23.2%	3.55	High delivery risk and weak satisfaction
SP → AL	256	22.7%	3.68	High-risk route with stronger volume
MA → SP	124	21.5%	3.92	High late rate, but moderate satisfaction
SP → MA	493	19.5%	3.68	High volume and weak satisfaction
SP → SE	208	18.1%	3.80	Elevated late rate and below-average review

Route hotspots. PR → AL is the most critical route in the available results: 32.6% of orders were late and its average review score was only 3.44. RJ → CE also stands out, with 23.2% late deliveries and a 3.55 review score. SP → AL and SP → MA are particularly important because they combine elevated late-delivery rates with substantially larger order volumes, making them potentially more impactful operationally than small-volume routes.

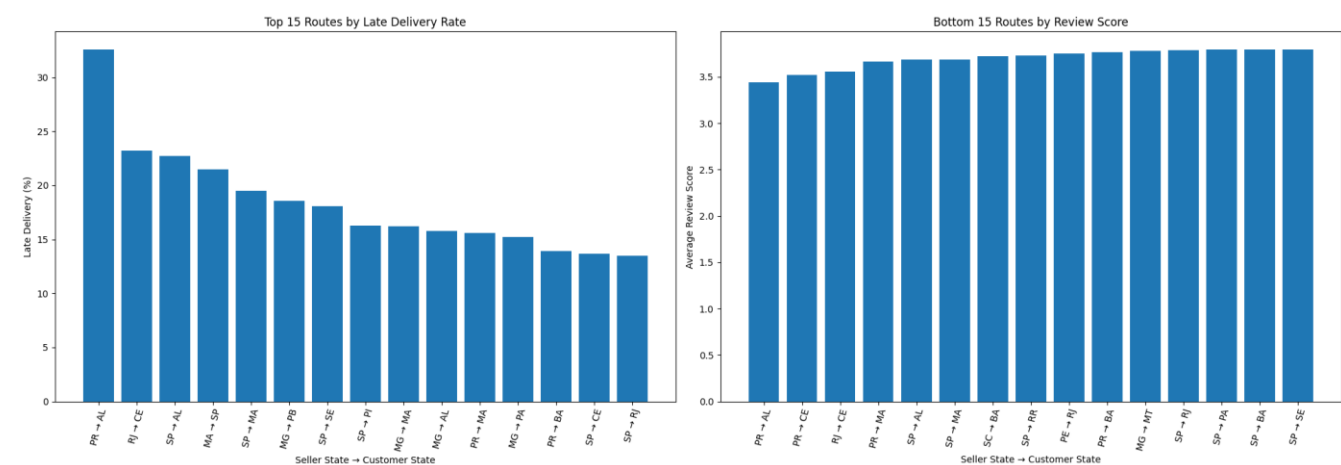


Figure 4. Top 15 routes by late-delivery rate and bottom 15 routes by average review score.

7. Recommendations

- Prioritize AL and MA for operational review. Investigate carrier performance, fulfillment lead times, distance, handoff delays and estimated-delivery accuracy for these destinations.
- Create route-level exception monitoring for PR → AL, RJ → CE, SP → AL and SP → MA, using both late-delivery percentage and review score.
- Separate high-risk/high-volume routes from high-risk/low-volume routes. High-volume routes such as SP → AL and SP → MA can generate a larger absolute number of dissatisfied customers.
- Use customer-state performance as an early-warning dashboard, but avoid treating state averages as proof of a geographic cause. Seller mix, product type, distance and carrier differences may contribute to the observed pattern.
- Track late-delivery rate and review score together. A route with both high lateness and low satisfaction is a stronger intervention candidate than a route that performs poorly on only one metric.
- Reassess the minimum-order thresholds periodically and consider confidence intervals or statistical tests in a future iteration to determine whether observed differences are statistically significant.

8. Conclusion

The analysis provides clear evidence of regional and route-level variation in delivery performance and customer satisfaction. AL and MA emerge as the most important state-level concerns, while PR → AL is the strongest route hotspot in the reported results. The overlap between high late-delivery rates and lower review scores suggests that delivery reliability is an important component of the customer experience. The next analytical step should be to identify the operational drivers behind these patterns particularly carrier, distance, seller and product effects, so that corrective actions can be targeted rather than based on geography alone.